ICT2217 Network Security

Lab Exercise 3.3: AAA Part 2 - Authorization

Learning Outcomes

Upon completion of this laboratory exercise, you should be able to:

- Configure authorization using privilege levels
- Configure authorization using role-based CLI views

Required Hardware

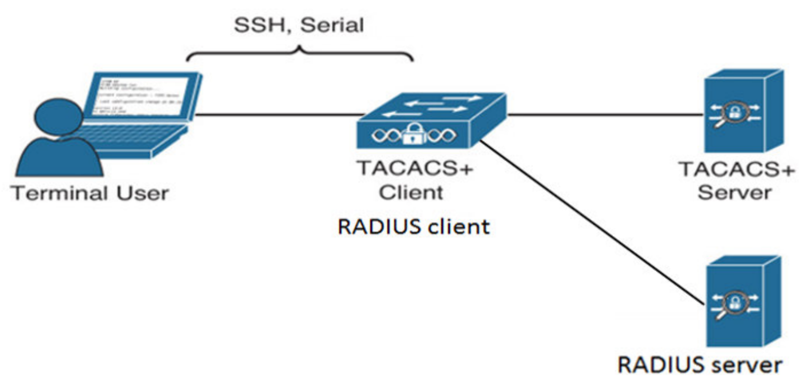
- 1 x Rack of Cisco networking devices
- 1 x Box of cables containing
  - USB-to-RJ45 console cables
  - Ethernet cables
- 3 x Laptops

Required Software

- Tera Term https://teratermproject.github.io/index-en.html
- Kali Linux Live Boot on USB drive
- TACACS.net 2.1.2 Demo version for Windows https://www.softpedia.com/get/Internet/Servers/Other-Servers/TACACS-net.shtml
- FreeRADIUS 3.2.3 for Kali Linux

PART 1: Configuring Authorization using Privilege Levels

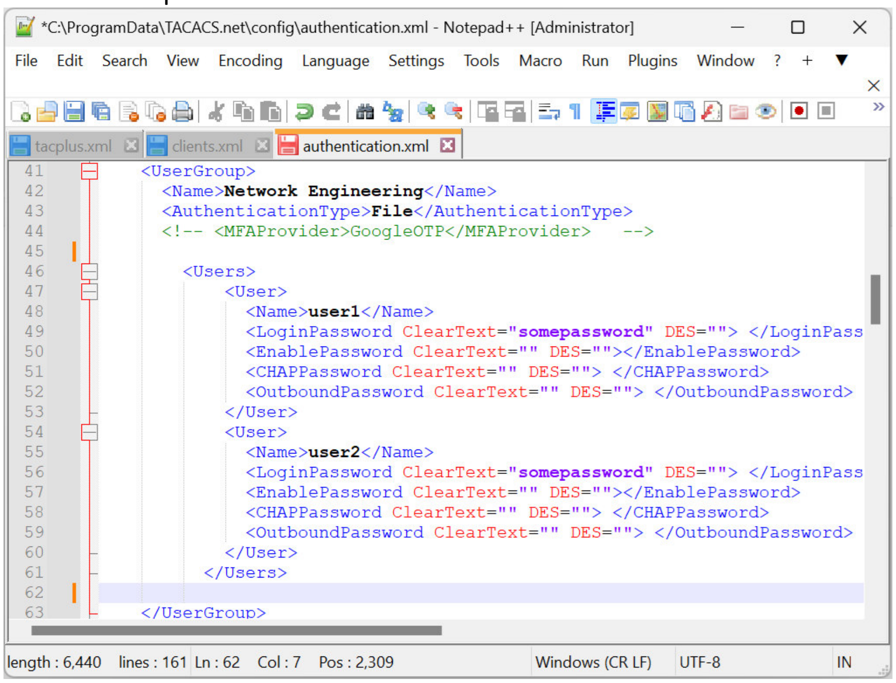
1.1 Note that AAA Authorization will only work when AAA Authentication is also configured. Hence, set up and re-configure the same network topology as done in Lab Exercise 3.2 Part 3.



1.2 Follow the example in Lab 3.3 Notes pg 5 to assign some commands of your choice to different privilege level.

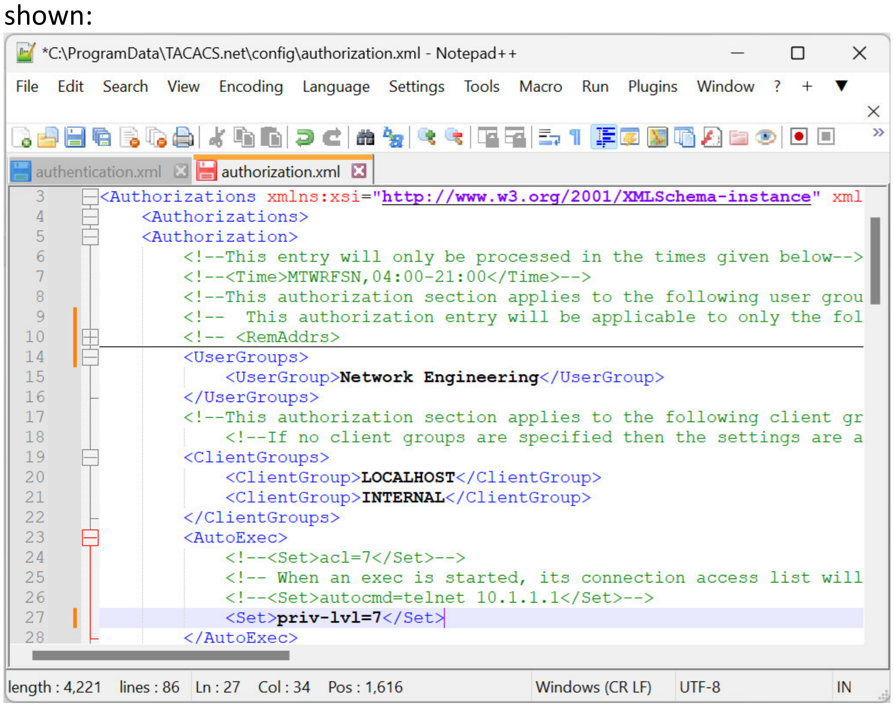
1.3 To assign privilege level to users locally on the network device, follow the examples in Lab 3.3 Notes pg 7.

1.4 To assign privilege level to users in centralised TACACS+ server, remember to first configure user authentication in the **authentication.xml** file as done in Lab Exercise 3.2 Part 2 re-produced here:



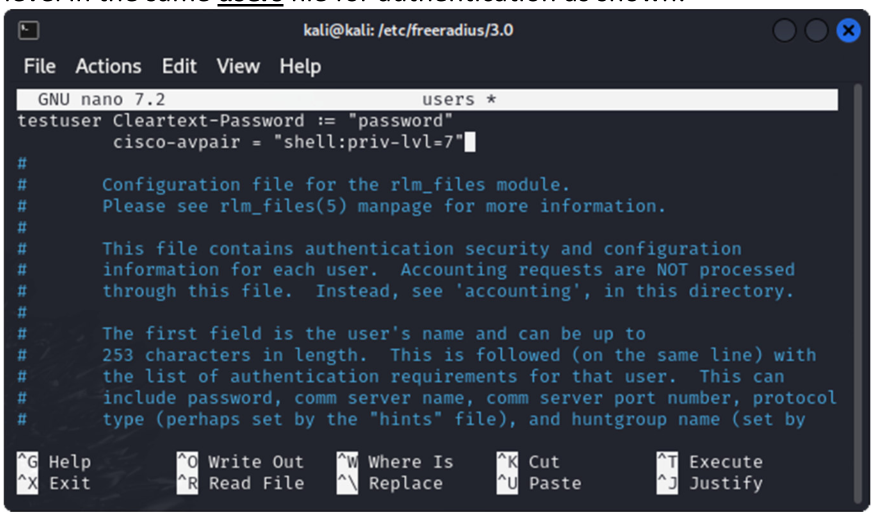

```
<UserGroup>
  <Name>Network Engineering</Name>
  <AuthenticationType>File</AuthenticationType>
  <!-- <MFAProvider>GoogleOTP</MFAProvider>   -->

    <Users>
        <User>
          <Name>user1</Name>
          <LoginPassword ClearText="somepassword" DES=""> </LoginPass
          <EnablePassword ClearText="" DES=""></EnablePassword>
          <CHAPPassword ClearText="" DES=""> </CHAPPassword>
          <OutboundPassword ClearText="" DES=""> </OutboundPassword>
        </User>
        <User>
          <Name>user2</Name>
          <LoginPassword ClearText="somepassword" DES=""> </LoginPass
          <EnablePassword ClearText="" DES=""></EnablePassword>
          <CHAPPassword ClearText="" DES=""> </CHAPPassword>
          <OutboundPassword ClearText="" DES=""> </OutboundPassword>
        </User>
      </Users>

</UserGroup>
```

Then open the corresponding **authorization.xml** file to assign the privilege level as shown:




```
<Authorizations xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xml
    <Authorizations>
    <Authorization>
        <!--This entry will only be processed in the times given below-->
        <!--<Time>MTWRFSN,04:00-21:00</Time>-->
        <!--This authorization section applies to the following user grou
        <!--  This authorization entry will be applicable to only the fol
        <!-- <RemAddrs>
        <UserGroups>
            <UserGroup>Network Engineering</UserGroup>
        </UserGroups>
        <!--This authorization section applies to the following client gr
            <!--If no client groups are specified then the settings are a
        <ClientGroups>
            <ClientGroup>LOCALHOST</ClientGroup>
            <ClientGroup>INTERNAL</ClientGroup>
        </ClientGroups>
        <AutoExec>
            <!--<Set>acl=7</Set>-->
            <!-- When an exec is started, its connection access list will
            <!--<Set>autocmd=telnet 10.1.1.1</Set>-->
            <Set>priv-lvl=7</Set>
        </AutoExec>
```

1.5 To assign privilege level to users in centralised RADIUS server, include the privilege level in the same **<u>users</u>** file for authentication as shown:



1.6 Finally, enable AAA authorization as shown in Lab 3.3 Notes pg 8.

1.7 For experiment, configure different privilege level for TACACS+ server, RADIUS server and local login.

1.8 When done, enter 'exit' until you are disconnected.

1.9 Now, login again using console connection or SSH. What is your privilege level after login?

1.10 Are you able to execute commands of the same or lower privilege levels?

1.11 Are you able to execute commands of higher privilege levels?

1.12 Simulate TACACS+ server is down by disconnecting it from the switch, then try login again. What is your privilege level after login?

1.13 Next, simulate RADIUS server is down also by disconnecting it from the switch, then try login again. What is your privilege level after login?

<u>PART 2: Configuring Authorization using Role-Based CLI Views</u>

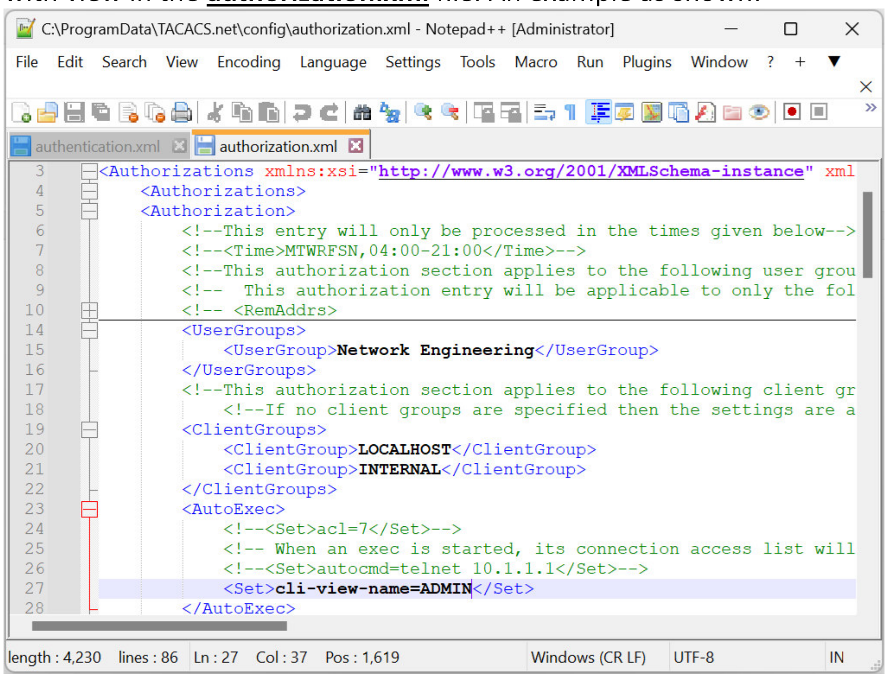
2.1 Continue with the same network topology in Part 1. Remember to connect back TACACS+ and RADIUS servers.

2.2 Follow the examples in Lab 3.3 Notes pg 11 - 14, create CLI views of your choice and configure some commands into them.

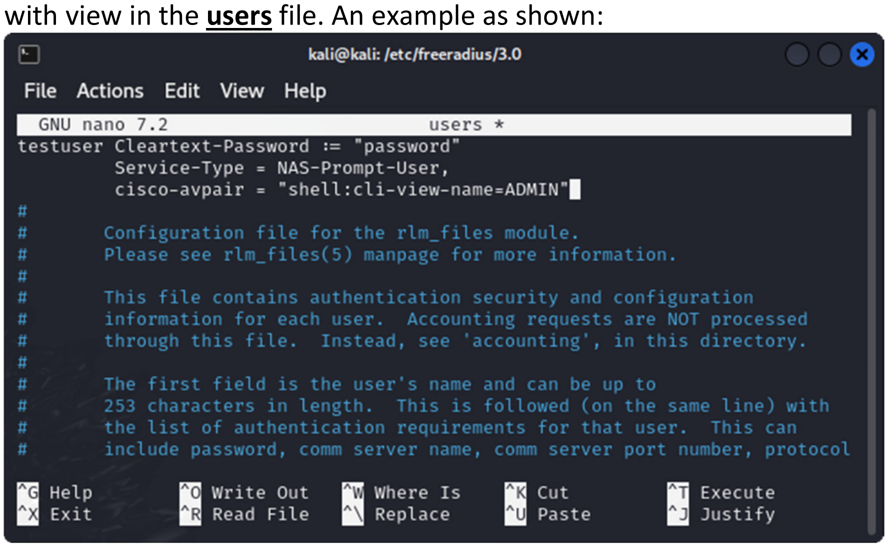
2.3 Next, follow the examples in Lab 3.3 Notes pg 15 - 16, create superviews of your choice and configure some CLI views that you've configured in Step 2.2 into them.

2.4 Finally, assign views to users as shown in Lab 3.3 Notes pg 18.

2.5 To assign views to users in the centralised TACACS+ server, **replace** privilege level with view in the **authorization.xml** file. An example as shown:



2.6 Similarly, to assign views to users in the centralised RADIUS server, **replace** priviledge with view in the **users** file. An example as shown:



2.7 For experiment, configure different views for TACACS+ server, RADIUS server and local login.

2.8 When done, enter 'exit' until you are disconnected.

2.9 Now, login again using console connection or SSH. What is your assigned view after login?

2.10 What are the commands available to you?

2.11 Simulate TACACS+ server is down by disconnecting it from the switch, then try login again. What is your assigned view after login?

2.12 Next, simulate RADIUS server is down also by disconnecting it from the switch, then try login again. What is your assigned view after login?